Equalities, Inequalities and Minimizers

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Equalities

Mastery of Elementary Algebra (a.k.a. "College Algebra") is a prerequisite for calculus, which is in turn a prerequisite for most statistics and data science courses. Nevertheless, each year, some students are still uncomfortable with algebraic manipulations of mathematical formulas. Therefore, I include this section as a quick reference.

Definition 0.1 (Transitive relation)**.** Let $\sim$ be a relation on a set S. The relation $\sim$ is **transitive** if, for all $a, b, c \in S$, $a \sim b$ and $b \sim c$ together imply $a \sim c$.

Example 0.1 (A transitive relation and one that is not)**.**

- $\leq$ is transitive: for example, $1 \leq 2$ and $2 \leq 5$, and indeed $1 \leq 5$.
- $\neq$ is not transitive: $1 \neq 2$ and $2 \neq 1$, but $1 \neq 1$ is false.

Theorem 0.1 (Equalities are transitive)**.** *If $a = b$ and $b = c$, then $a = c$. That is, $=$ is a transitive relation.*

Theorem 0.2 (Substituting equivalent expressions)**.** *If $a = b$, then for any function $f(x)$, $f(a) = f(b)$*

Definition 0.2 (Identity)**.** An **identity** is an equation between two expressions that holds for every value of its variables in a stated set, such as the real numbers.

Example 0.2 (An identity, and an equation that is not one)**.**

- $2(x + 1) = 2x + 2$ is an identity: it holds for every real x. For example, at $x = 3$, both sides equal 8.

- $x + 1 = 3$ is not an identity: it holds at $x = 2$, but at $x = 0$ the left side is 1 and the right side is 3.

Inequalities

Definition 0.3 (Strict and non-strict inequalities)**.** For real numbers a and b, $a < b$ ("a is less than b") and $a > b$ ("a is greater than b") are **strict inequalities**. The **non-strict inequalities** also allow equality:

- $a \leq b$ means $a < b$ or $a = b$;
- $a \geq b$ means $a > b$ or $a = b$.

Example 0.3 (Strict and non-strict inequalities)**.**

- $3 \leq 3$ is true, because $3 = 3$, but $3 < 3$ is false.
- $2 < 3$ and $2 \leq 3$ are both true.
- $4 \geq 5$ is false, because $4 > 5$ and $4 = 5$ are both false.

Theorem 0.3 (Adding to both sides of an inequality)**.** *If $a < b$, then $a + c < b + c$*

Theorem 0.4 (Negating both sides of an inequality)**.** *If $a < b$, then: $-a > -b$*

Theorem 0.5 (Multiplying both sides of an inequality by a positive number)**.** *If $a < b$ and $c > 0$, then $ca < cb$.*

Theorem 0.6 (Negation is multiplication by -1)**.**

$$-a = (-1) * a$$

Powers, square roots, and absolute values

Definition 0.4 (Power, base, and exponent)**.** Let a be a real number and n a natural number. The nth **power** of a is the product of n copies of a:

$$a^n \stackrel{\text{def}}{=} \underbrace{a \cdot a \cdot \cdots \cdot a}_{n \text{ factors}}$$

In a^n, a is the **base** and n is the **exponent**. The second power a^2 is the **square** of a.

Example 0.4 (Powers)**.**

- $$\begin{aligned} 2^3 &= 2 \cdot 2 \cdot 2 \\ &= 8, \end{aligned}$$

  with base 2 and exponent 3.

- $$\begin{aligned} (-3)^2 &= (-3) \cdot (-3) \\ &= 9, \end{aligned}$$

  but

  $$\begin{aligned} -3^2 &= -(3^2) \\ &= -9 : \end{aligned}$$

  the exponent applies only to the 3.

- $5^0 = 1$.

- $$\begin{aligned} 2^{-2} &= \frac{1}{2^2} \\ &= \frac{1}{4}. \end{aligned}$$

Definition 0.5 (Powers with exponent zero or a negative integer)**.** For the exponent 0 and negative integer exponents:

- $a^0 \stackrel{\text{def}}{=} 1$, including $0^0 = 1$ by convention;
- $a^{-n} \stackrel{\text{def}}{=} \frac{1}{a^n}$, when $a \neq 0$.

Remark 0.1 (Powers with real exponents). Definition 14 in Polynomials, Exponentials and Logarithms extends powers to real exponents when the base is positive.

Definition 0.6 (Square root)**.** For a real number $a \geq 0$, the **square root** of a, written $\sqrt{a}$, is the non-negative real number whose square (Definition 0.4) is a.

Example 0.5 (Square roots)**.**

- $\sqrt{9} = 3$. Although $(-3)^2 = 9$ too, -3 is negative, so $\sqrt{9}$ is not -3.
- $\sqrt{0} = 0$.
- $\sqrt{2} \approx 1.41421$.
- $\sqrt{-4}$ is not a real number, because no real number squares to -4 (see Section 1 in Complex Numbers).

Definition 0.7 (Absolute value of a real number)**.** The **absolute value** of a real number a is

$$|a| \stackrel{\text{def}}{=} \begin{cases} a, & a \geq 0 \\ -a, & a < 0 \end{cases}$$

Example 0.6 (Absolute values)**.**

- $|2.5| = 2.5$, because $2.5 \geq 0$.
- $$\begin{aligned} |-3| &= -(-3) \\ &= 3, \end{aligned}$$

  because $-3 < 0$.
- $|0| = 0$.
- $$\begin{aligned} |-3| &= 3 \\ &= \sqrt{9} \\ &= \sqrt{(-3)^2}\,: \end{aligned}$$

  for every real a, $|a| = \sqrt{a^2}$ (Definition 0.6).
- $|x - 1| < 1$ says that x is less than 1 away from 1, that is, $0 < x < 2$.

Remark 0.2 (Absolute value as distance). For real numbers a and b, $|a - b|$ is the distance between a and b on the number line.

Minimum, maximum, argmin and argmax

Exercise 0.1 (Smallest value, and where it occurs)**.**

1. Let $f(x) = (x-2)^2 + 1$ for every $x \in \mathbb{R}$. What is the smallest value f takes, and at which x does it take that value?
2. Now let $g(x) = (x-2)^2 + 1$ only for $x \in \{0, 1, 3, 4\}$. What is the smallest value g takes, and at which x does it take that value?

Solution

Solution 0.1.

1. The square of a real number is never negative, so $(x-2)^2 \geq 0$ for every x, and therefore $f(x) \geq 0 + 1 = 1$. The square is 0 only when $x - 2 = 0$, that is, when $x = 2$, and there $f(2) = 1$. So the smallest value of f is 1, and $x = 2$ is the only input where f takes it.

2. Evaluate g at each of its four inputs:

x	0	1	3	4
$g(x)$	5	2	2	5

   The smallest value of g is 2, and g takes it at two inputs, $x = 1$ and $x = 3$. The answer to "where?" is the set $\{1, 3\}$, not a single number.

Definition 0.8 (Minimum)**.** Let $A \subseteq \mathbb{R}$. A number m is the **minimum** of A, written $\min A$, if $m \in A$ and $m \leq a$ for all $a \in A$.

Remark 0.3 (Not every set has a minimum). The set $\{3, 1, 4\}$ has minimum 1, and the interval $[0, 1]$ has minimum 0. The interval $(0, 1]$ has no minimum: every element a of it has a smaller element, $a/2$, also in it.

Definition 0.9 (Maximum)**.** Let $A \subseteq \mathbb{R}$. A number M is the **maximum** of A, written $\max A$, if $M \in A$ and $a \leq M$ for all $a \in A$.

Example 0.7 (A maximum, and a set without one)**.**

- $\max\{-2, 5, 0\} = 5$: 5 is in the set, and $-2 \leq 5$, $5 \leq 5$, $0 \leq 5$.
- The interval $[0, 1)$ has no maximum. For any $a \in [0, 1)$, the number $\frac{a+1}{2}$ is also in $[0, 1)$, and it is larger than a, since $\frac{a+1}{2} > a$ exactly when $a < 1$. So no element of $[0, 1)$ is at least as large as every element.

Definition 0.10 (Argmin)**.** Let $f : A \to \mathbb{R}$ be a function. The **argmin** of f over A is the set of inputs where f takes its smallest value:

$$\arg\min_{x \in A} f(x) \stackrel{\text{def}}{=} \{x \in A : \forall a \in A,\ f(x) \leq f(a)\}$$

When this set has exactly one element $\hat{x}$, we write $\hat{x} = \arg\min_{x \in A} f(x)$.

Remark 0.4 (Smallest value versus where it occurs). The smallest value itself is $\min f(A)$, the minimum (Definition 0.8) of the image of f, and the argmin is where that value is attained. For example, let $f(x) = x^2$ on $A = \{-1, 0, 2\}$. The image is $f(A) = \{1, 0, 4\}$, so the smallest value is $\min f(A) = 0$, and $\arg\min_{x \in A} f(x) = \{0\}$.
The argmin is empty when f has no smallest value, for example $f(x) = x$ on $(0, 1]$.

Definition 0.11 (Argmax)**.** Let $f : A \to \mathbb{R}$ be a function. The **argmax** of f over A is the set of inputs where f takes its largest value:

$$\arg\max_{x \in A} f(x) \stackrel{\text{def}}{=} \{x \in A : \forall a \in A,\ f(a) \leq f(x)\}$$

When this set has exactly one element $\hat{x}$, we write $\hat{x} = \arg\max_{x \in A} f(x)$.

Example 0.8 (An argmax with two points, and an empty one)**.**

- Let $g(x) = x^2$ on $A = \{-2, 0, 1, 2\}$. The values are $g(-2) = 4$, $g(0) = 0$, $g(1) = 1$ and $g(2) = 4$, so the largest value is 4 and $\arg\max_{x \in A} g(x) = \{-2, 2\}$. This argmax has two elements, so it is not written as a single $\hat{x}$.
- Let $f(x) = x$ on $[0, 1)$. As in Example 0.7, for every input x the input $\frac{x+1}{2}$ has a larger value, so no input attains a largest value, and $\arg\max_{x \in [0,1)} f(x)$ is the empty set.

Global and local minimizers

Exercise 0.2 (A dip that is not the bottom)**.** Let $f(x) = x^3 - 3x$ for $x \in \mathbb{R}$.

1. Factor $f(x) - f(1)$ completely.
2. Use part 1 to compare $f(x)$ with $f(1)$ for every x in the interval $(0, 2)$. Is $f(x)$ larger than, smaller than, or equal to $f(1)$, and for which x?
3. Compute $f(-3)$. Is $f(1)$ the smallest value f takes on all of $\mathbb{R}$?
4. Does f take a smallest value anywhere on $\mathbb{R}$?

Solution

Solution 0.2.

1. Since

$$\begin{aligned} f(1) &= 1 - 3 \\ &= -2, \end{aligned}$$

so

$$\begin{aligned} f(x) - f(1) &= x^3 - 3x + 2 \\ &= (x-1)(x^2 + x - 2) \\ &= (x-1)(x-1)(x+2) \\ &= (x-1)^2(x+2). \end{aligned}$$

2. For $x \in (0, 2)$, $(x-1)^2 \geq 0$ and $x + 2 > 0$, so their product is at least 0. By part 1, $f(x) - f(1) \geq 0$, that is, $f(x) \geq f(1)$. So $f(x)$ is never smaller than $f(1)$ on $(0, 2)$. It equals $f(1)$ only when $(x-1)^2 = 0$, that is, only at $x = 1$, and it is larger than $f(1)$ at every other $x \in (0, 2)$.

3. $$\begin{aligned} f(-3) &= -27 + 9 \\ &= -18. \end{aligned}$$

   Since $-18 < -2 = f(1)$, $f(1)$ is not the smallest value of f on $\mathbb{R}$.

4. No. For each $n \geq 2$,

$$\begin{aligned} f(-n) &= -n^3 + 3n \\ &= -n(n^2 - 3) \\ &\leq -n, \end{aligned}$$

so no value of f is smaller than all the others: for any x, picking $n \geq 2$ with $-n < f(x)$ gives $f(-n) < f(x)$.

Definition 0.12 (Global minimizer)**.** Let $A \subseteq \mathbb{R}^p$ and let $f : A \to \mathbb{R}$ be a function. A point $x^* \in A$ is a **global minimizer** of f over A if $f(x^*) \leq f(x)$ for all $x \in A$.

Remark 0.5 (Global minimizers form the argmin). The global minimizers of f are exactly the elements of $\arg\min_{x \in A} f(x)$ (Definition 0.10). A function can have more than one global minimizer. For example, $f(x) = (x^2 - 1)^2$ on $\mathbb{R}$ satisfies $f(x) \geq 0$ for every x, and $f(x) = 0$ exactly when $x = -1$ or $x = 1$, so its global minimizers are -1 and 1, and $\arg\min_{x \in \mathbb{R}} f(x) = \{-1, 1\}$.

Definition 0.13 (Local minimizer)**.** Let $A \subseteq \mathbb{R}^p$ and let $f : A \to \mathbb{R}$ be a function. A point $x^* \in A$ is a **local minimizer** of f if there is a number $\delta > 0$ such that $f(x^*) \leq f(x)$ for all $x \in A$ with $\|x - x^*\| < \delta$, where $\|\cdot\|$ is the Euclidean norm.

Remark 0.6 (Local and global minimizers). For $p = 1$, $\|x - x^*\| = |x - x^*|$. For example, with $x^* = 1$ and $\delta = 1$, the condition $|x - 1| < 1$ means $0 < x < 2$.
Every global minimizer (Definition 0.12) is a local minimizer: take any $\delta > 0$. For example, $x^* = 0$ is a global minimizer of $f(x) = x^2$ on $\mathbb{R}$, so it is also a local minimizer.
The converse fails. In Exercise 0.2, $x^* = 1$ is a local minimizer of $f(x) = x^3 - 3x$ (take $\delta = 1$), but not a global one.

Example 0.9 (A point that is not a local minimizer)**.** For $f(x) = x^3 - 3x$, the point $x = -1$ is not a local minimizer. For any t,

$$\begin{aligned}
f(-1+t) &= (-1+t)^3 - 3(-1+t) && \text{(substitute)} \\
&= (-1 + 3t - 3t^2 + t^3) - 3(-1+t) && \text{(expand the cube)} \\
&= -1 + 3t - 3t^2 + t^3 + 3 - 3t && \text{(multiply } -3 \text{ into the parentheses)} \\
&= 2 - 3t^2 + t^3 && (3t - 3t = 0 \text{ and } -1 + 3 = 2) \\
&= 2 + t^2(t-3) && \text{(factor out } t^2) \\
&= f(-1) + t^2(t-3), && (f(-1) = (-1)^3 - 3(-1), \text{ which equals } 2)
\end{aligned}$$

and $t^2(t-3) < 0$ whenever $t \neq 0$ and $t < 3$. So whatever $\delta > 0$ is, the point $x = -1 + t$ with $t = \min\{\delta/2, 1\}$ satisfies $|x - (-1)| < \delta$ and $f(x) < f(-1)$. For example,

$$\begin{aligned} f(-0.9) &= -0.729 + 2.7 \\ &= 1.971 \\ &< 2. \end{aligned}$$